Evolutionary Robotics
2: Biological Fundamentals

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Outline

Introduction
Why Evolutionary Robotics?

Biological fundamentals
Natural Selection, neural Systems

Evolutionary algorithms
Selection, Mutation, Crossover

Artificial neural networks
Controller for robots

How to evolve robots
(Encoding, Fitness, Simulation)

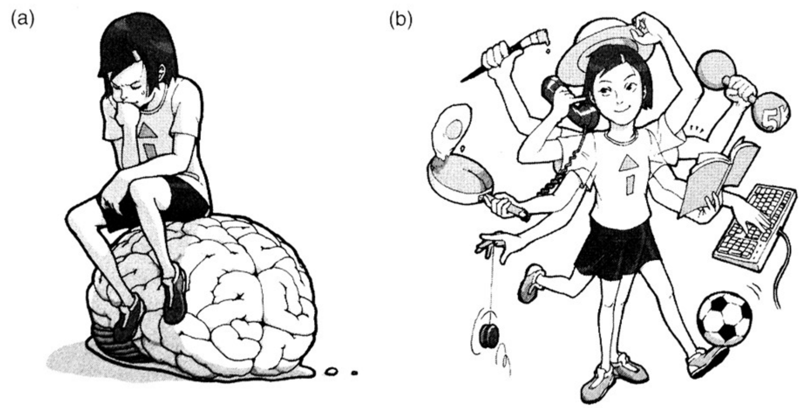
Evolution of simple navigation
(Phototaxis, Obstacle avoidance)

Power and Limitations of Reactive Intelligence

Evolving Morphology
(Body + Brain Co-evolution)

Challenges and new approaches
Reality Gap, Diversity, DNNs

Recap (1/2)



"Cogito ergo sum."

Computer science:

input first, then processing, then output

Embodied cognition:

- body is relevant, interaction with the environment based on feedbacks
- "outsourcing calculations to the body/environment"

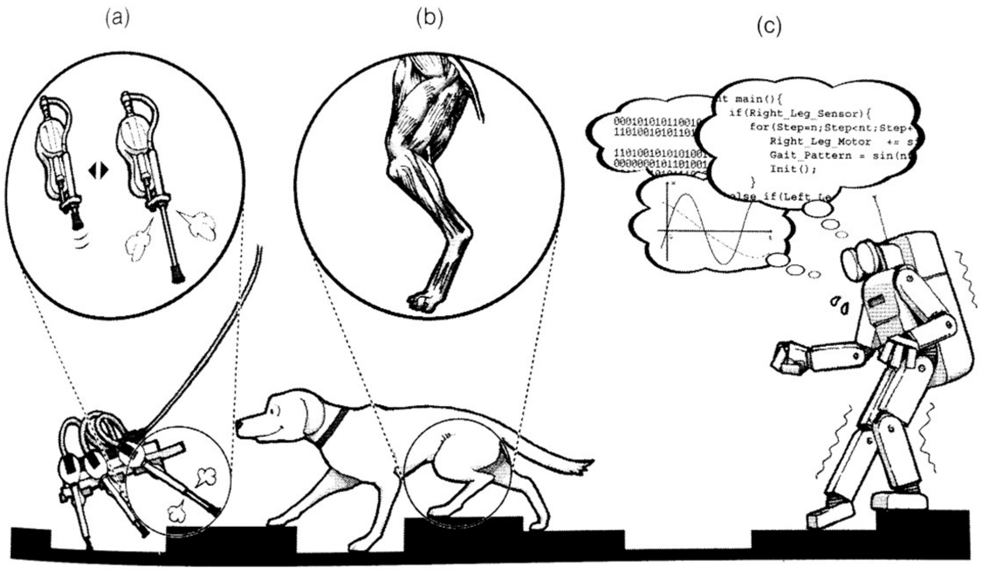
from: Rolf Pfeifer and Josh C. Bongard. How the body shapes the way we think - A new view of intelligence. MIT Press, November 2006.

Recap (2/2)

(a) robot exploiting material properties of its legs

(b) animal exploiting material properties of its legs

(c) robot built from stiff materials must apply complex control



from: Rolf Pfeifer and Josh C. Bongard. How the body shapes the way we think - A new view of intelligence. MIT Press, November 2006.

Embodied cognition

"The aim is not to create a brain simulation so detailed and accurate [. . .] Instead, the aim is to **capture enough** of the computationally functional properties of the brain to enable the resultant emulation to perform intellectual work [. . .] much of the **messy biological detail of a real brain is irrelevant**."

– Superintelligence, Bostrom

a lot about

emulating brains,

quantification of required calculation time, etc.

But

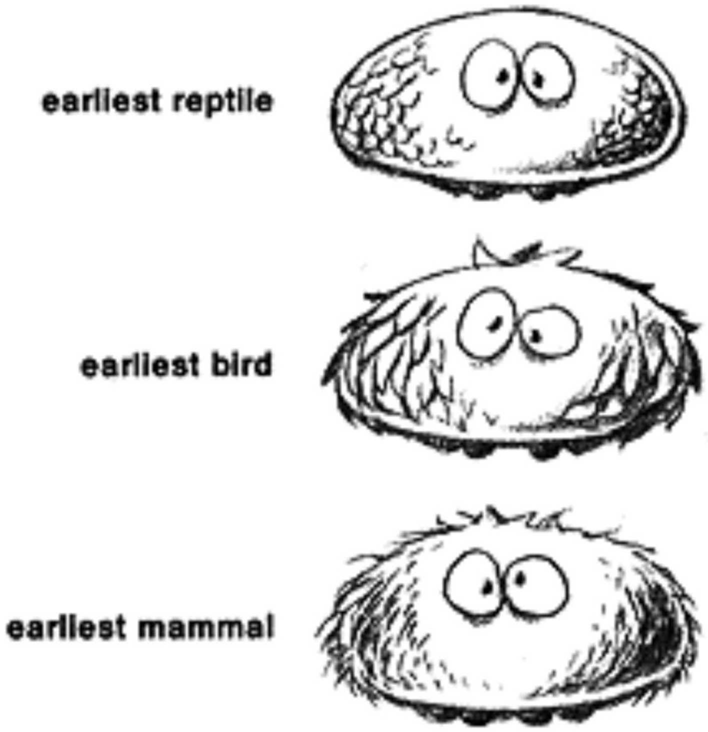
qualities of the evolutionary process?

environment? embodiment?

Evolution

2: Biological Fundamentals

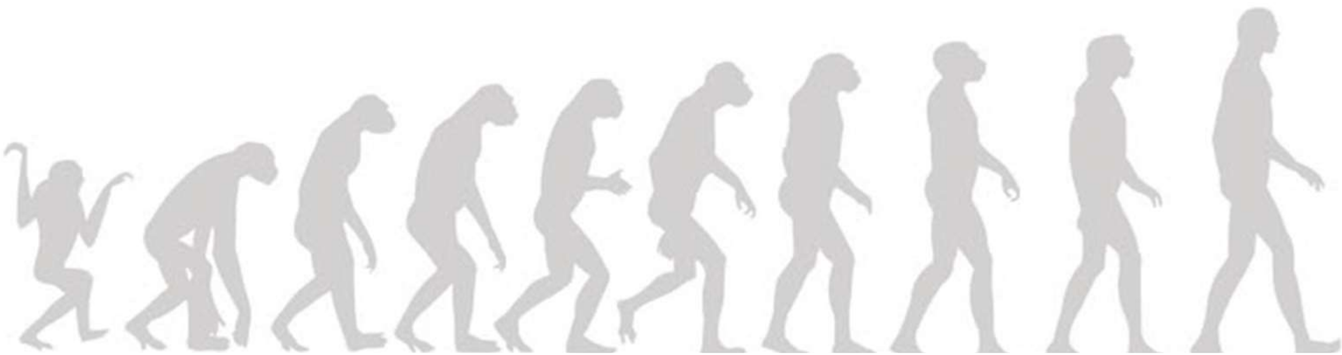
Theory to explain where all species come from



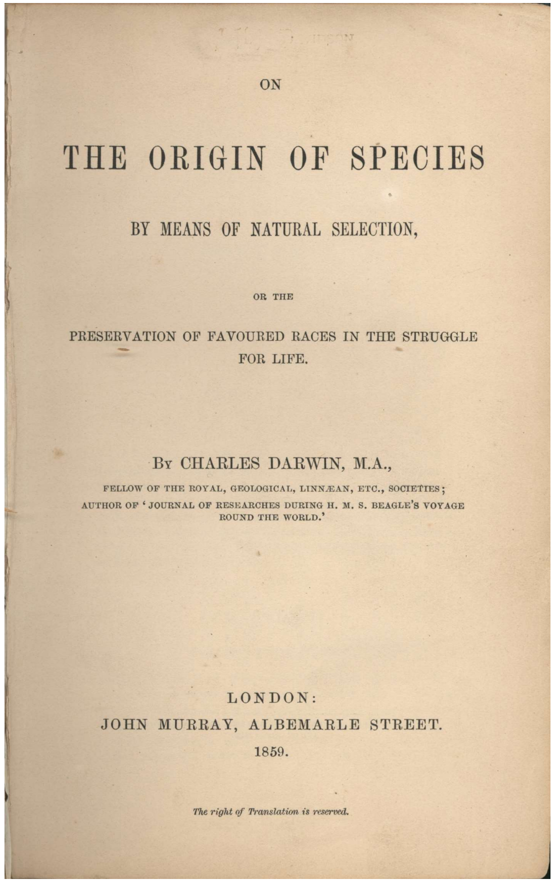
Why bother with natural evolution?

Why should we learn about natural evolution given that we plan to study artificial evolution as a process to generate robot behaviors?

- It is important to know the differences between natural and artificial evolution and their shared characteristics.
- The initial inspiration of artificial evolution was biological and future improvements might also be bio-inspired.
- Besides, the more you know the more creative you can be.



From Darwin to automated DNA sequencing



ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE FOR LIFE.

BY CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE ROUND THE WORLD.'

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

The right of Translation is reserved.

“Theory of evolution”

Evolution of species by
natural selection

by Charles Darwin (1809-1882), published in 1858
and his book *Origin of species* in 1859

Alfred Russel Wallace had the same idea but wrote a letter to Darwin. . .

Either way the idea didn’t come out of nothing:
Jean-Baptiste Lamarck, Geoffroy Saint-Hilaire, Erasmus Darwin (grandfather of Charles), Robert Edmond Grant, Robert Chambers

Preparatory work by Malthus

An Essay on the Principle of Population, published in 1798 by Thomas Robert Malthus (1766-1834)

Main Message:
continued population growth drives a society towards the fundamental problem of overstretching its resource limitations and results are poverty, epidemics, famines, or wars

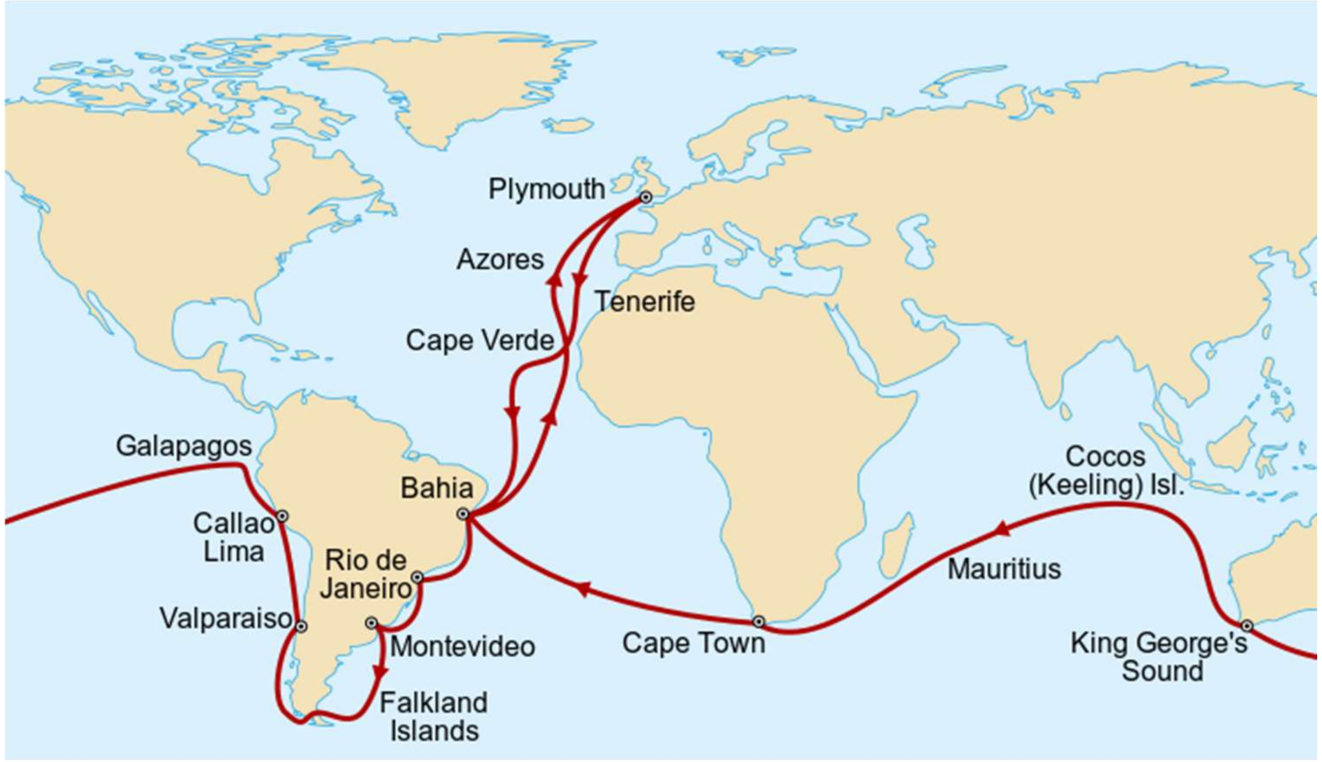
Voyage of the HMS Beagle

27 December 1831 - 2 October 1836

key events:

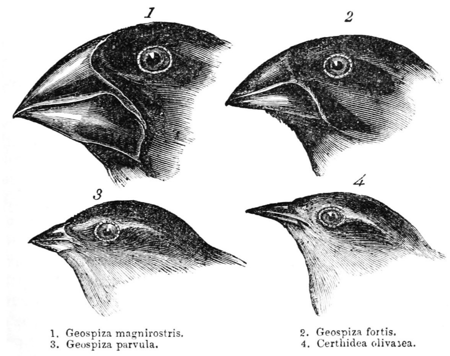
- geological phenomena
- differences between

species on Galápagos Islands



Galápagos Islands, Ecuador

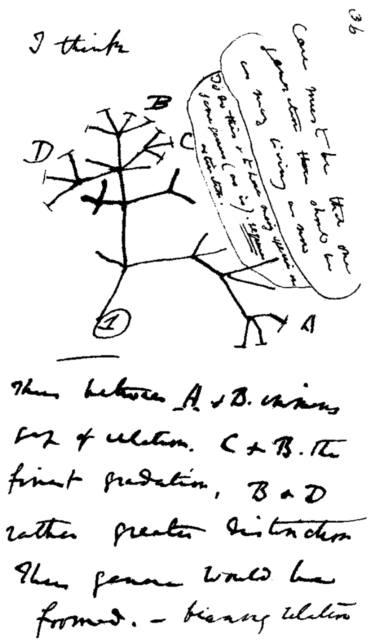
Finches and an evolutionary tree



Differences in the beaks of finches on the Galápagos Islands

Hypothesis:
all evolved from the same species
that made its way to the islands (distance: 926 km)

Darwin starts a notebook in 1837, p. 36 shows a schematic tree



Public response to Darwin

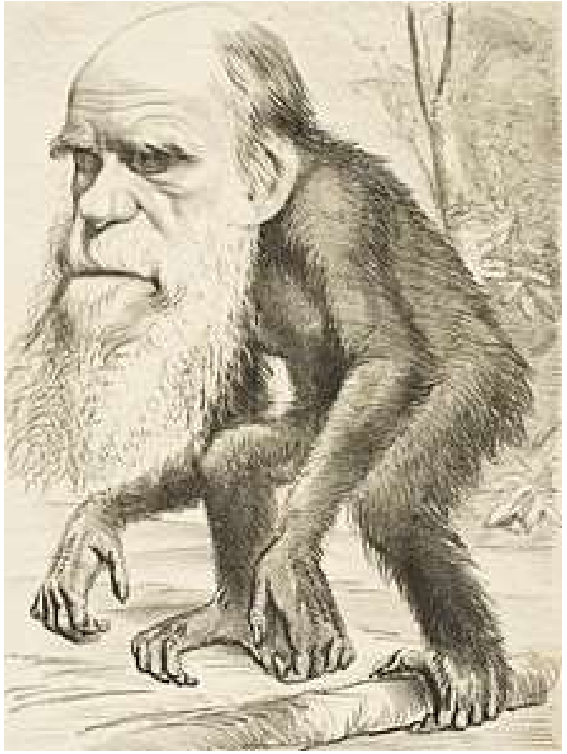
Darwin had hesitated to publish for long and the response was as expected

(criticism, mockery, and disgust).

But his theory was also appreciated by many.

Unfortunate progress of history:
Social Darwinism, eugenics, racism, imperialism, etc.

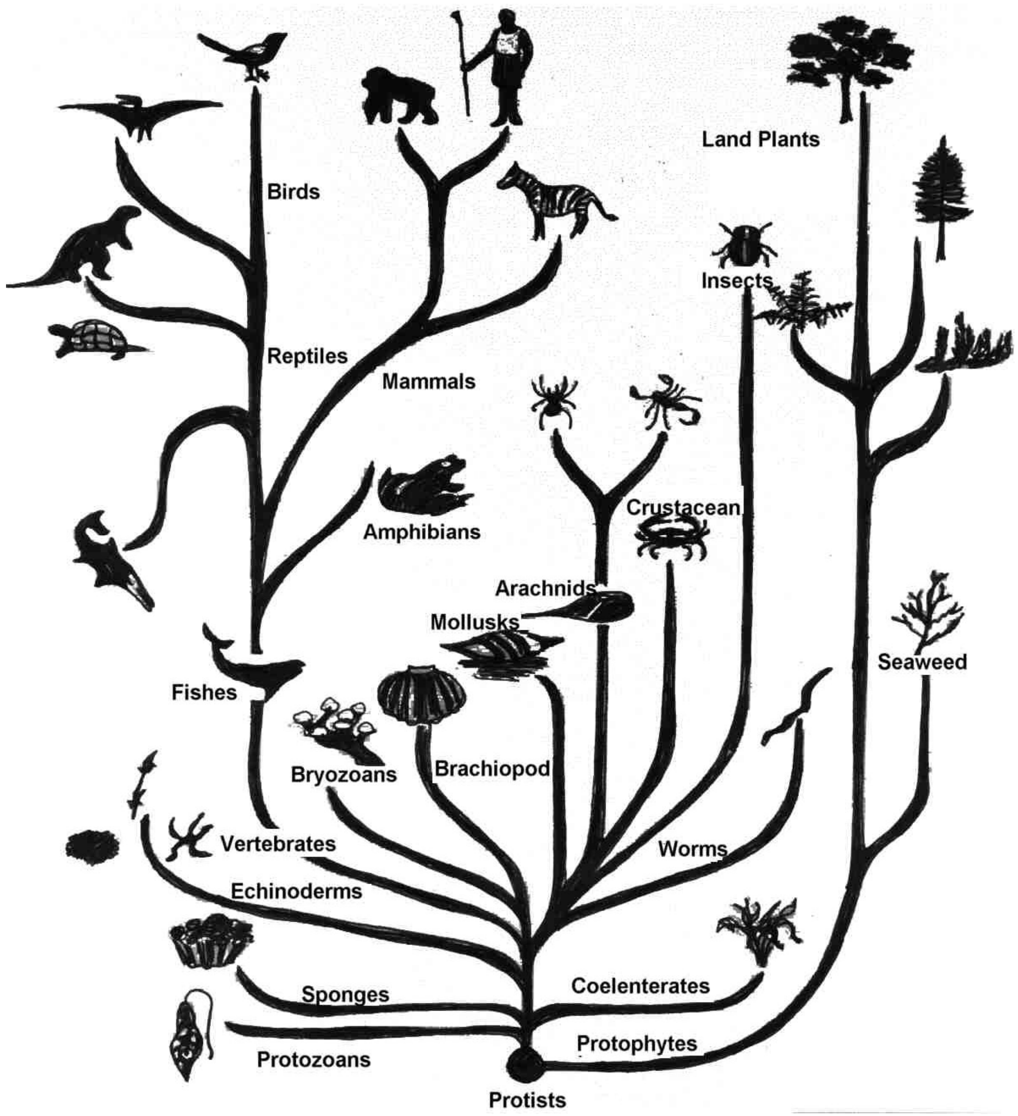
Even today: Intelligent Design and creationism. . .



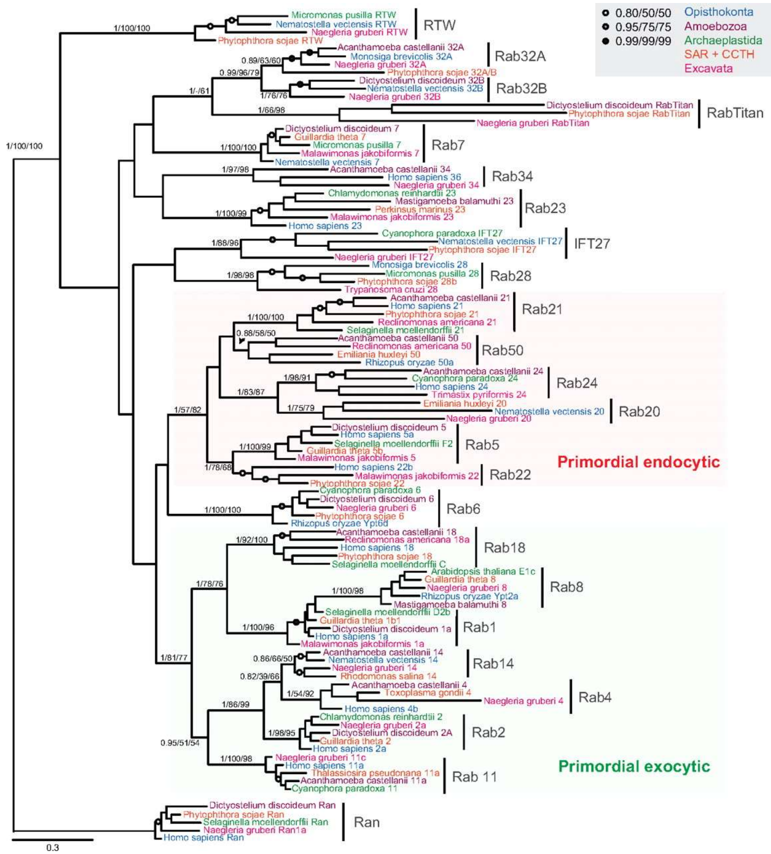
(caricature, 1871)

Phylogenetic tree

Inferred evolutionary relationships between species



Computational phylogenetics



First sequenced DNA:

bacteriophage Φ-X174 in 1977

Computational phylogenetics

- subfield of Bioinformatics
- application of computational methods to Phylogenetics

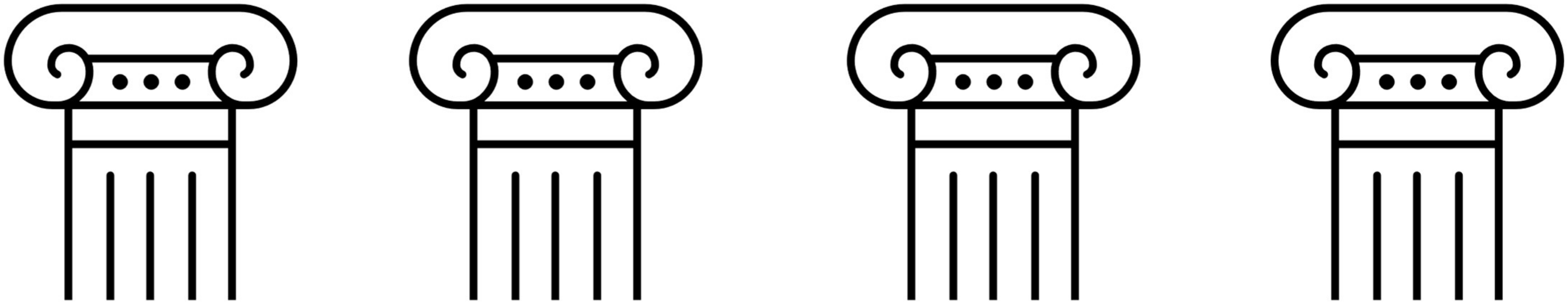
Idea: Phylogenetics as optimization problem, what is the most likely tree?

Fundamentals of Evolution

2: Biological Fundamentals

The 4 pillars of evolution

Population **Diversity** **Heredity** **Selection**



“minimal requirements of evolution”

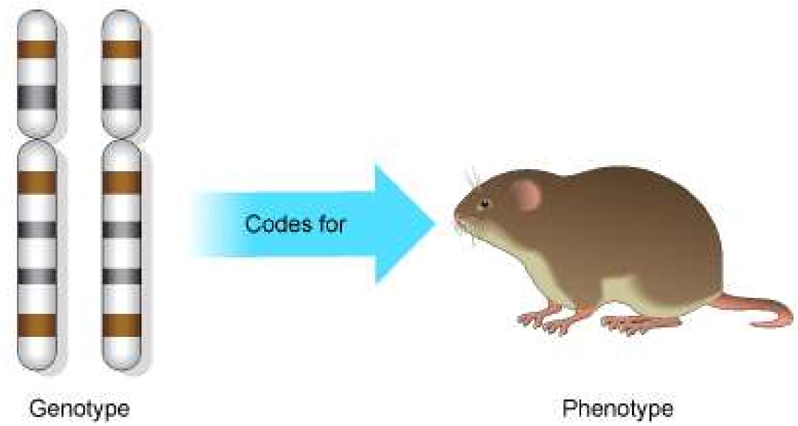
The 4 pillars of evolution

Population : group of several individuals
Diversity: individuals have different characteristics
Heredity : characteristics are transmitted over generations
Selection : individuals make more offspring than the environment can support
⇒ organisms are competing for limited resources

Better at food gathering ⇒ better at surviving ⇒ make more offspring

(How to implement that for artificial evolution?)

Phenotype & Genotype



Genotype : the genetic material of an organism
selection does not operate directly on it,
it is affected by mutations

Phenotype: the actual organism (appearance, behavior, etc.)
is observable,
selection operates on it,
it is affected by environment, development, and learning

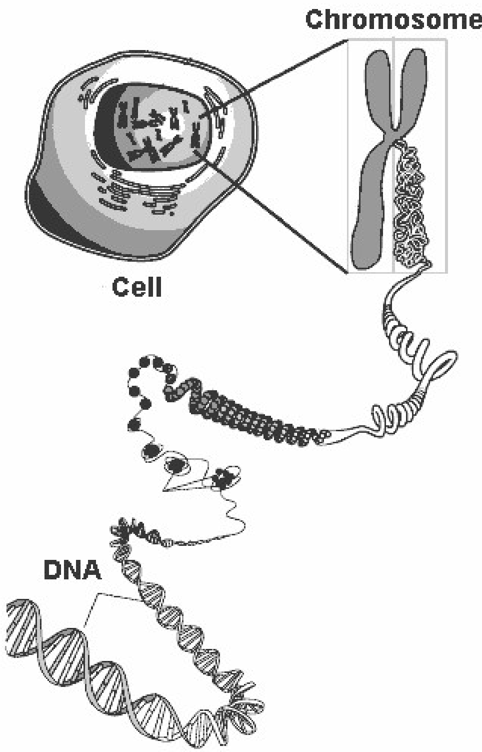
DNA

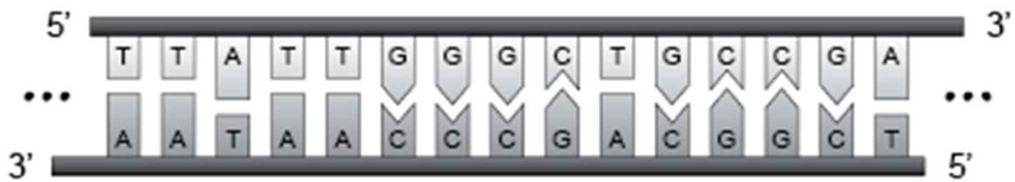
DeoxyriboNucleic Acid, long molecule, twisted in spiral, and compressed

Humans have 46 chromosomes, structured in 23 pairs (diploid)

(respectively one from the mother and one from the father)

DNA is composed of 2 complementary sequences (strands) of 4 nucleotides
(Adenine, Thymine, Cytosine, Guanine), which bind in pairs
(A-T and C-G)

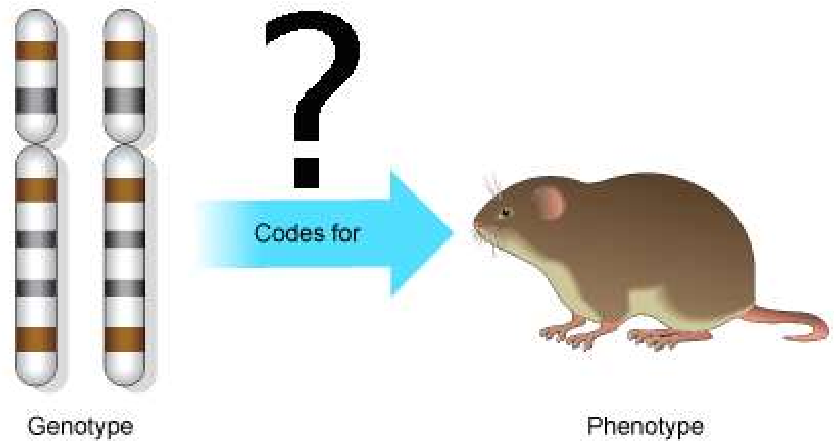




From DNA to 'function'

The long way of gene expression (example of eukaryotic genes):

1. Transcription: DNA to RNA (ribonucleic acid)
2. RNA processing
3. RNA export (from nucleus to cytoplasm, sometimes towed by motor proteins)
4. Translation: creation of a protein
5. Protein folding

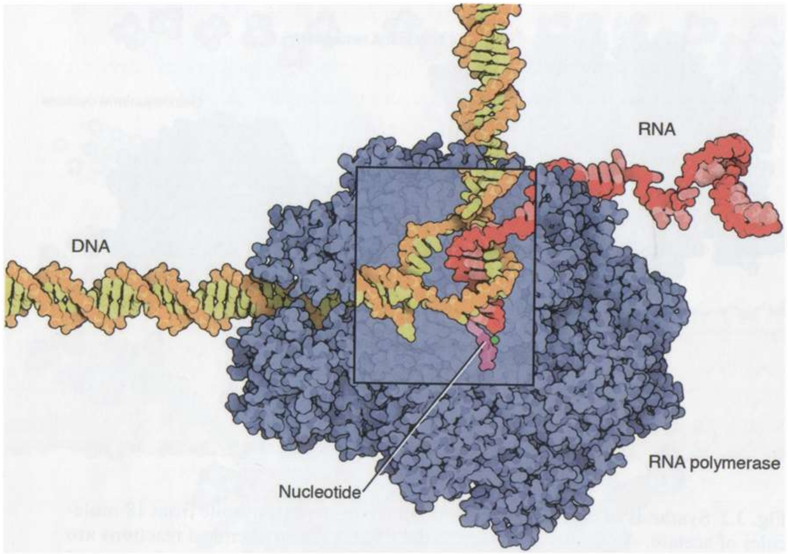


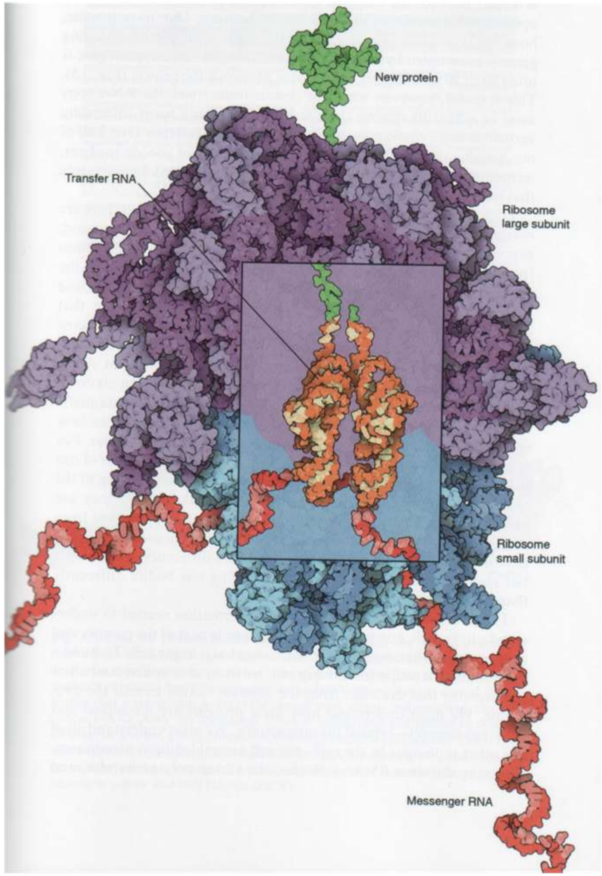
Transcription

RNA polymerase is an enzyme

⇒ a catalyst

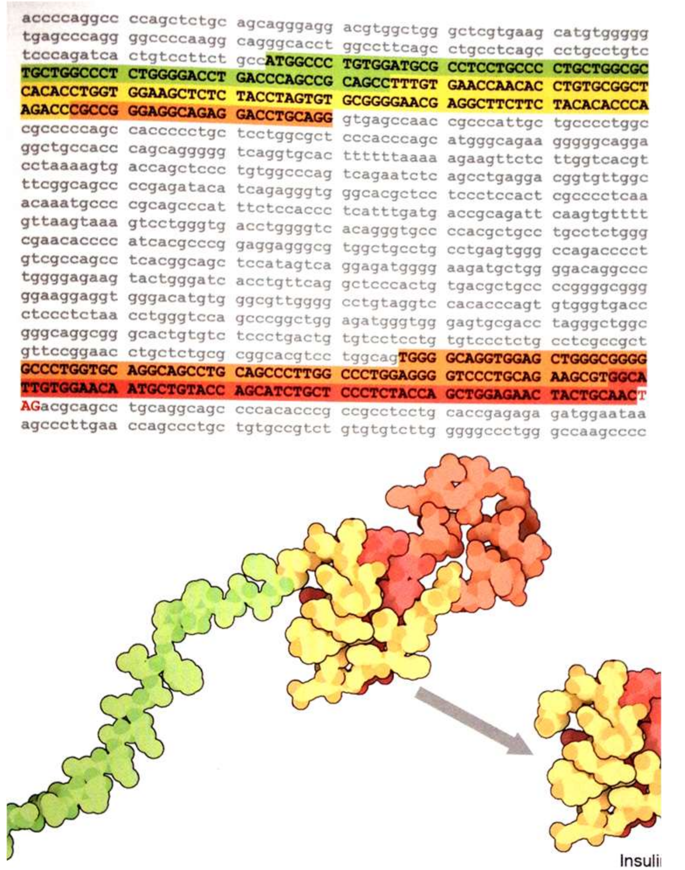
⇒ speeds up chemical reactions





Protein folding – insulin

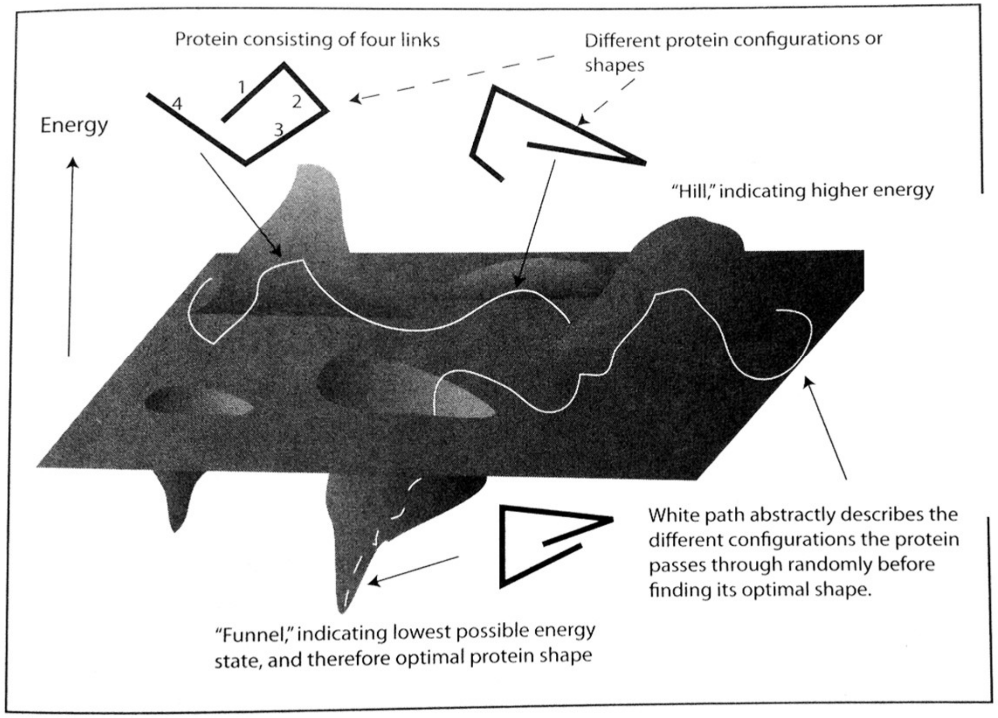
Gene for human insulin: A short segment from the human genome is shown here, including the gene that encodes insulin. Several layers of processing must occur to create the mature form of the protein. The protein-encoding sequence is broken into two segments in the DNA, shown here in capital letters. Notice that it starts with start codon A-T-G and ends with the stop codon T-A-G. After the messenger RNA is transcribed and spliced to connect these two segments, a long protein with 98 amino acids is synthesized. This is then trimmed, removing the signal sequence at the start (shown in green) and a loop in the middle (shown in orange). The two remaining pieces form the mature protein.



(from David S. Goodsell, The machinery of life, Springer)

Protein folding – energy landscape (1/2)

protein structure prediction is very difficult
(⇒ molecular dynamics simulations, 2013 Nobel Prize in Chemistry)



(Peter M. Hoffmann, Life's Ratchet, Basic Books, 2012)

Off-topic(?): Water on Mars

"Why do we think aliens are made of water?

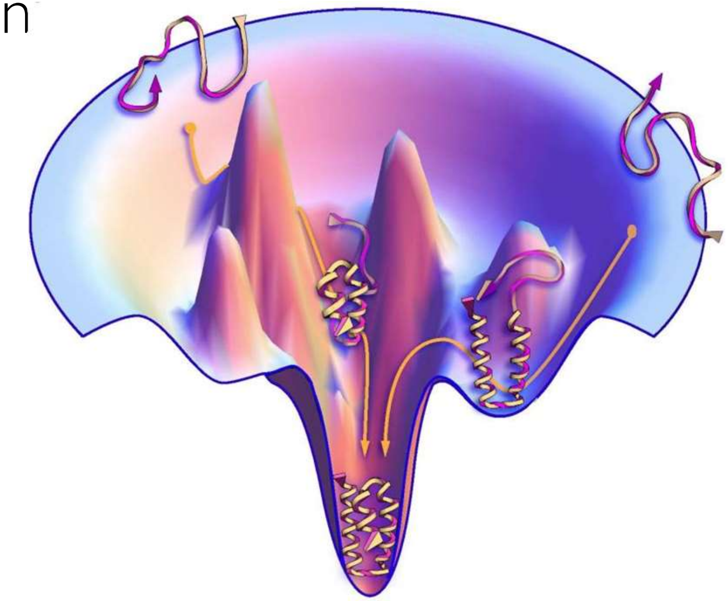
What makes water so useful? First of all, it serves as a substrate for all the chemical reactions you need to make a living thing.

To get something as complicated as biology, you've got to have a system that allows a wide variety of molecules to interact in a wide variety of ways. Water, which is a polar molecule–i.e., it has both positively and negatively-charged ends–acts as a 'universal solvent.' That means it can dissolve many chemicals–including the organic compounds that are the building blocks of life on Earth–and allow them to recombine or attach to one another in various arrangements."

(from: http://www.slate.com/articles/news_and_politics/explainer/2006/03/why_do_we_think_aliensare_made_of_water.html)

Protein folding – energy landscape (2/2)

Minimizing free energy, lowest-energy configuration



Hydrophobic amino acids are safely tucked inside,
hydrophilic amino acids are outside

Technical terms and concepts

2: Biological Fundamentals

Important technical terms and concepts

1. Mutation
2. Recombination
3. natural and sexual selection
4. misconceptions: fitness and the survival of the fittest
5. genetic drift
6. co-evolution
7. Speciation
8. Pleiotropy
9. Gene knockout

Mutation

There is a highly efficient DNA repair mechanism in each cell

Average **mutation rate** in humans:

$\sim \mathbf{10^{-10}}$ per base pair,
$\sim$ 0.1 per genome per generation

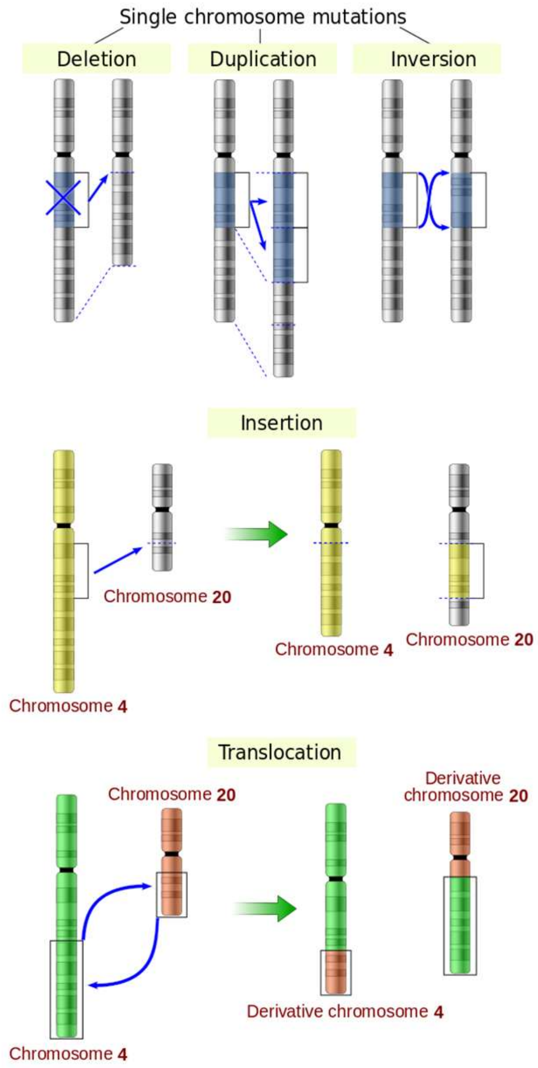
(very rough estimate)

mutation: unrepaired damage to DNA due to radiation, errors of copying etc.

point mutation: exchange of a single nucleotide for another

other types of mutations: deletion, duplication, inversion, insertion, translocation

Q: What are the consequences for the methods of computational phylogenetics?



Effects of mutations

The distribution of fitness effects caused by single-nucleotide substitutions in an RNA virus
https://doi.org/10.1073/pnas.040014610
Rafael Sanjuán, Andrés Moya, and Santiago F. Elena

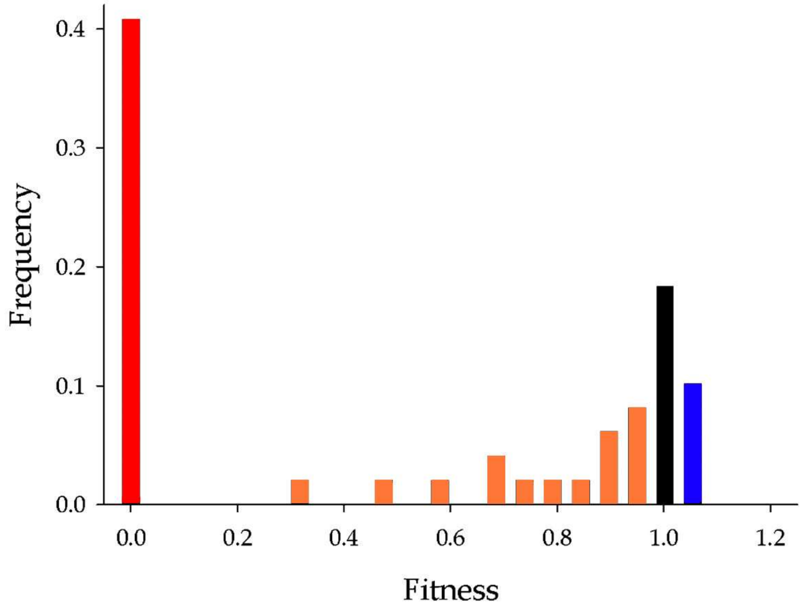
Fitness: success in reproduction

beneficial mutations increase fitness

neutral mutations have no effect (cf. genetic drift)

harmful mutations decreases the fitness of the organism

Lethal mutations cause death of the organism



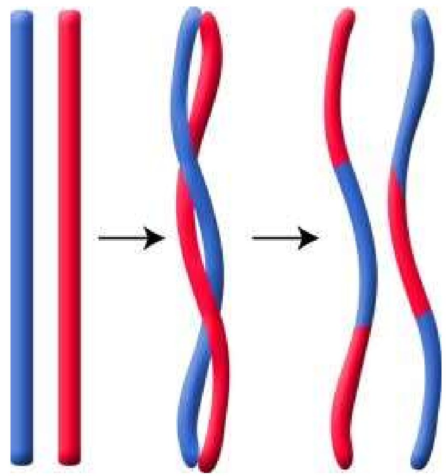
The distribution of fitness effects caused by single-nucleotide substitutions in an RNA virus

Recombination

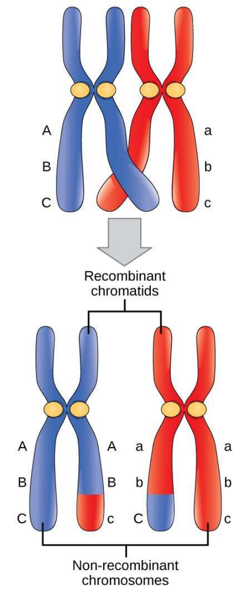
Major engine of genetic variation

Occurs under many circumstances, here we focus on:

Meiosis: special kind of cell division for sexual reproduction, chromosome set is reduced from **diploid** to **haploid**, during fertilization two haploid chromosome sets fuse to form a diploid set again

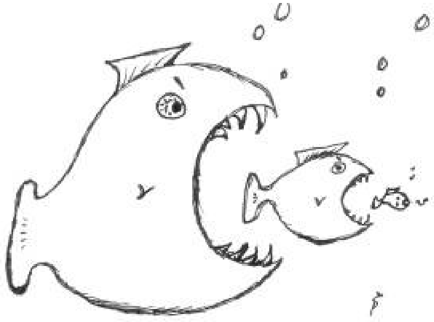


2-point crossover



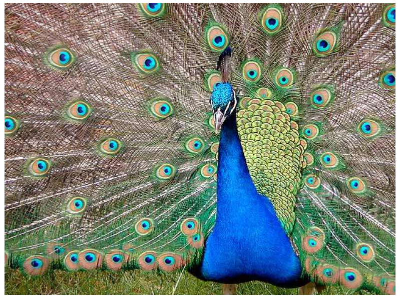
1-point crossover

Selection



natural selection

gradual process that changes the frequency of biological traits in a population based on reproductive success of organisms



sexual selection

organisms control reproductive access by choosing among a population of available mates

artificial selection

selective breeding, humans breed other animals and plants for particular traits

Misconceptions – fitness

Although fitness is assigned to individuals in evolutionary algorithms, it is actually not as explicit in natural evolution as one might think.

“fitness is a property, not of an individual, but of a class of individuals” (John Maynard Smith)

Fitness should be understood as a probability to survive and to have a certain number of offspring; hence it gets explicit only a posteriori

Misconceptions – survival of the fittest

"Survival of the fittest" was coined not by C. Darwin but by Herbert Spencer and it is misleading; natural selection is more about reproduction than survival (survival is only a prerequisite).

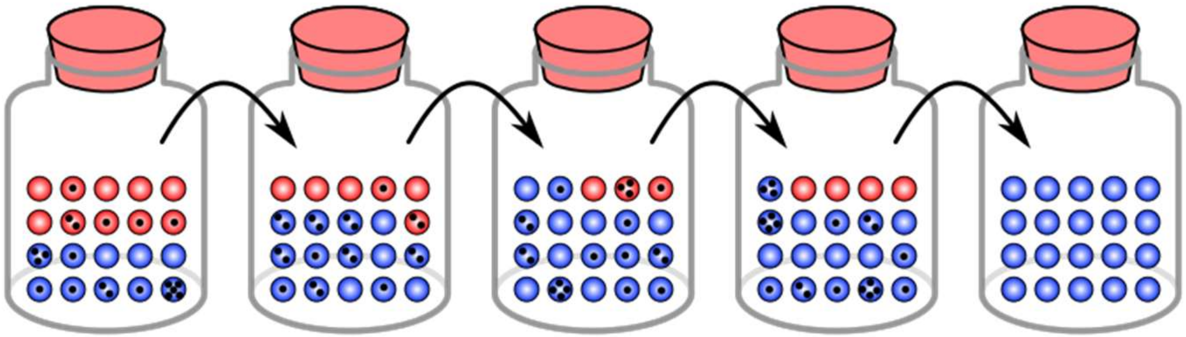
Natural selection does not grant organisms what they "need".

'Fit enough' instead of 'fittest' would be better because natural evolution is not identical to an optimizer, survival is not directly bound to optimality.

It can be misunderstood as a tautology "survival of those who are better equipped for surviving" but the actual concept is about being 'better adapted' and there is an independent concept of fitness

Genetic drift

Change in the frequency of a gene in a population due to **random sampling**, that is, without selective pressure



Similar to a random walk on a plateau (≙neutral region) of the fitness landscape

⇒ cf. neutral theory of molecular evolution

Co-evolution

Co-evolution is the change of an organism triggered by the change of a related organism

Examples: predator and prey (arms race!), host and parasite, pollination

Systems of co-evolution can be very complicated and can be difficult to understand.

It is frequent in natural evolution but less frequently used in artificial evolution.

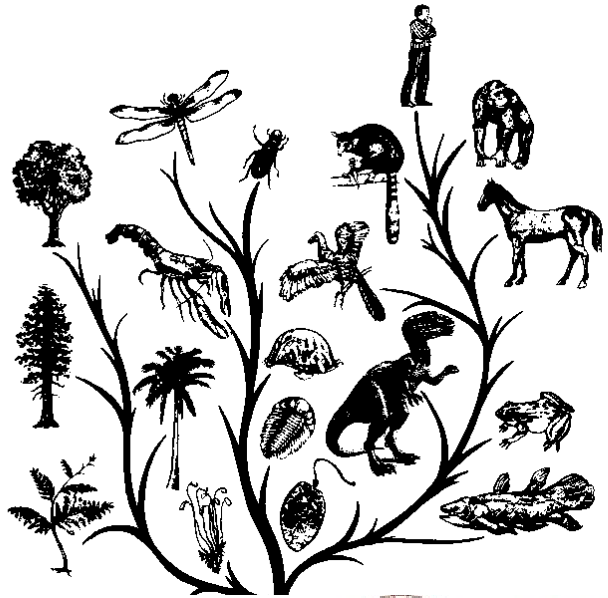
Speciation

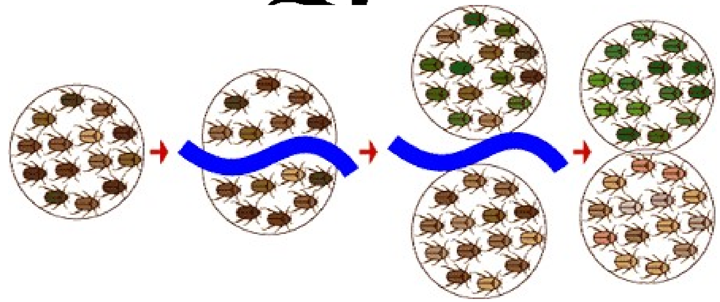
Speciation is the evolutionary process which generates new species

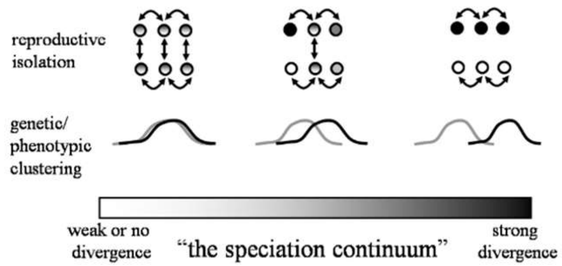
species: group of organisms capable of interbreeding

There are many theories on speciation and different types.

Example causes for speciation are: barrier formation (mountains, islands), entering new niches, genetic drift etc.

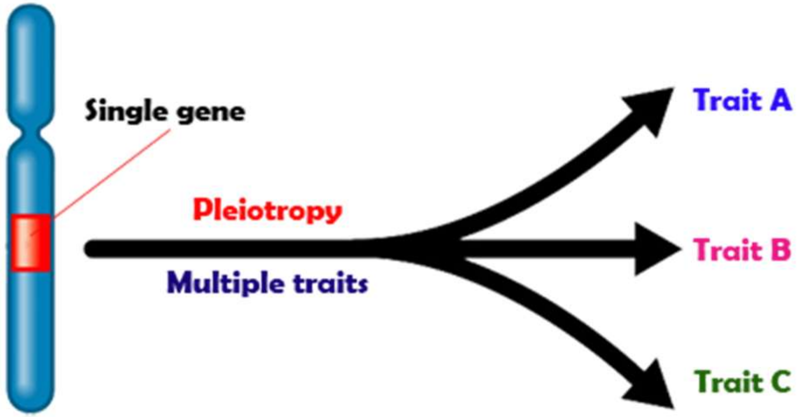






Pleiotropy

Pleiotropy: a gene has multiple effects on the phenotype, possibly over a wide range of seemingly unrelated features



For example, Gregor Mendel reported 'trait 3' in peas: brown seed coat, violet flowers, and axial spots

Gene Knockout

Gene knockout: method from genetics to determine the function of a gene; the investigated gene is fully deactivated and the result is observed

Challenges: redundancy & robustness

The knocked out gene might be replaced by the (self-organized) activation of other genes; as a consequence no effect is observed and cause & effect are difficult to determine

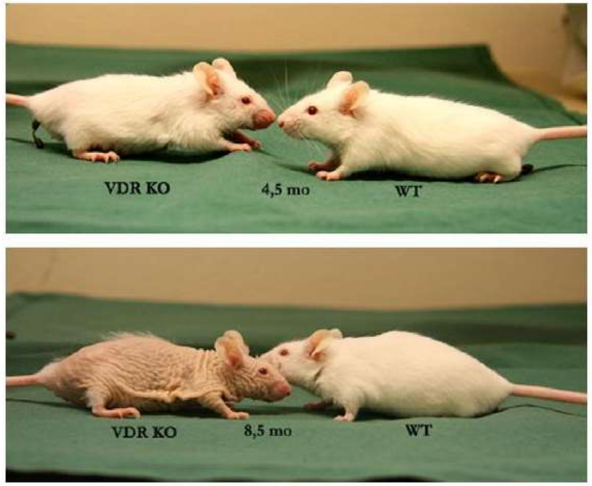


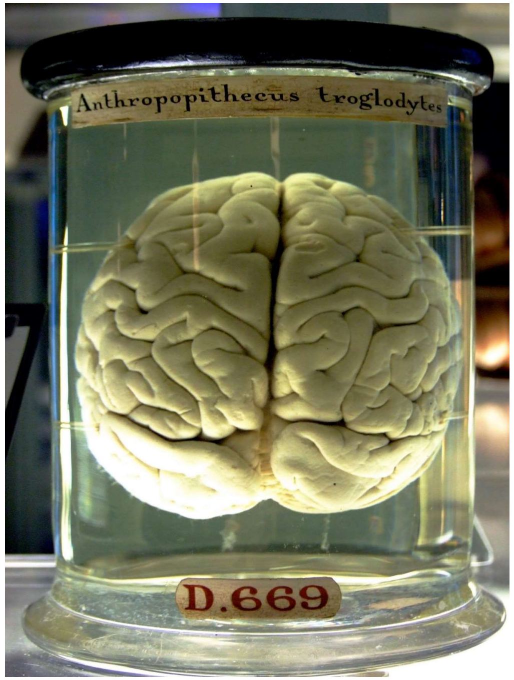
Fig. 2. Phenotype of VDR knockout mouse (KO) compared to wildtype littermate (WT; NMRI background strain) at the age of 4.5 (top) and 8.5 (bottom) months.

Keisala et al. Premature aging in vitamin D receptor mutant mice. J. Steroid Biochem. Mol. Biol. 2009 Jul;115(3-5):91-7

Neurobiology

2: Biological Fundamentals

Neurobiology and the brain



Neurobiology studies the nervous system

The **brain** is the center of the nervous system

Organisms are exposed to combinations of **information** from different sensory modalities (light, sound, touch etc.), the **brain interprets** these sensations on a more abstract level; it plans and executes motor behavior

Brains

Basic building block of the brain: neuron

Human brain has about 10^{11} neurons

Composition: cell body, dendrites, axon

Connection between neurons:

axon connects via synapse to dendrite

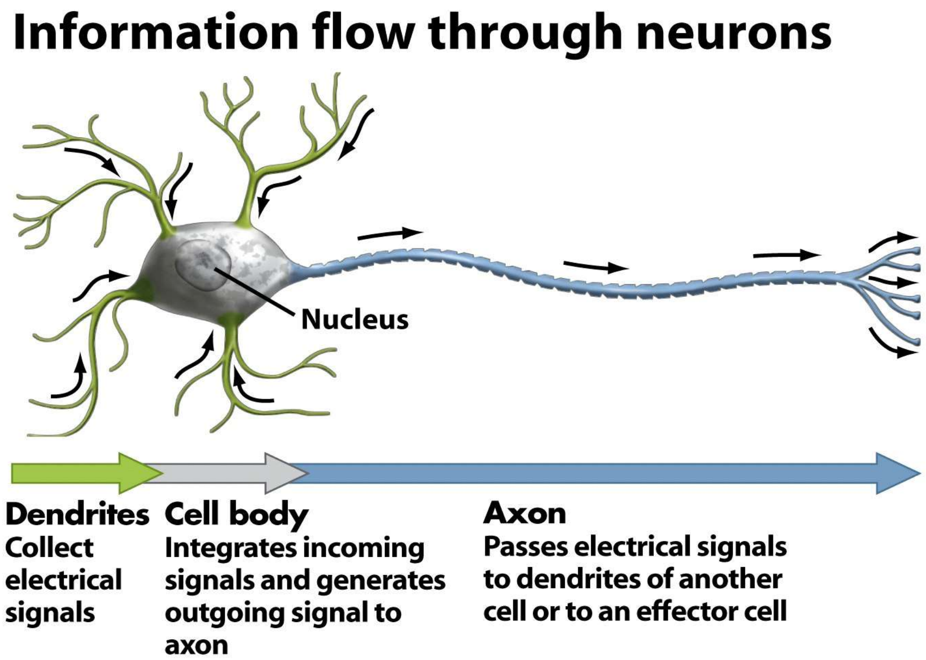


Figure 45-2b Biological Science, 2/e

Control concepts besides brains

note, there are organisms/machines showing ‘intelligent’ behavior without brains, for example:

unicellular organisms

passive dynamic walkers

References

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